Dimensionality Reduction & Unsupervised Clustering

PCA • K-Means • DBSCAN • Hierarchical Clustering

Project report | Scikit-learn Wine dataset (178 observations, 13 features)

1. Executive Summary

This project standardizes the 13 numerical Wine dataset features, applies PCA, and compares K-Means, DBSCAN, and Ward hierarchical clustering. 10 principal components retain 96.17% variance. Among k=2..10, K-Means achieves its highest silhouette at k=3 (0.299).

2. Objectives

Standardize features; perform PCA; plot explained variance and choose components; use K-Means with elbow and silhouette analysis; apply DBSCAN and hierarchical clustering; visualize clusters in 2D and 3D with Seaborn/Plotly; interpret and compare results.

3. Dataset & Preprocessing

The Scikit-learn Wine dataset has 178 observations and 13 continuous numerical features. Its three reference classes are excluded from model fitting and used only for post-hoc external validation. The notebook checks missing values, median-imputes defensively, and standardizes each feature with z-scores. Scaling is important because PCA and Euclidean distance are sensitive to feature magnitude.

4. PCA

PCA is fitted to standardized features. The smallest component count reaching 95% cumulative explained variance is 10 (96.17%). 8 components reach 90%. The scree plot reports individual and cumulative variance.

5. K-Means

K-Means is evaluated for k=2..10 on the first 10 PCs with k-means++ initialization, n_init=20 and random_state=42. Inertia is used for the elbow method; silhouette measures cohesion and separation. The maximum tested silhouette occurs at k=3, score=0.299. Internal metrics: Davies–Bouldin=1.336; Calinski–Harabasz=76.2.

6. DBSCAN

DBSCAN is tested at eps values 1.0–3.0 with min_samples=5. At eps=1.4, it finds 4 clusters excluding noise; 88.8% of samples are noise. Non-noise silhouette=0.557. The high noise fraction illustrates why silhouette alone is insufficient.

7. Hierarchical Clustering

Agglomerative clustering with Ward linkage is cut at k=3. Ward linkage minimizes the increase in within-cluster variance at each merge. The notebook includes a dendrogram to inspect merge structure. Post-hoc ARI: K-Means=0.897; hierarchical=0.759.

8. Visualization

2D and 3D plots show K-Means labels in PCA coordinates. The notebook also provides interactive Plotly versions. PCA plots are projections and may conceal separation in omitted components.

9. Comparison

K-Means creates a complete partition and works best for compact, roughly spherical groups. DBSCAN can discover dense, irregular groups and identify noise, but is sensitive to eps and density variation. Hierarchical clustering exposes nested structure, but results depend on linkage and cut level.

10. Conclusion

The workflow retains at least 95% of variance in 10 PCs and evaluates all three clustering families. Elbow and silhouette diagnostics support a K-Means candidate; DBSCAN demonstrates density-based clustering and the coverage/noise trade-off; hierarchical clustering provides a nested alternative.

11. Limitations & Future Work

Results are dataset-specific. The 95% threshold is a heuristic. Distance metric, PCA representation and hyperparameters influence outcomes. DBSCAN can be tuned further using k-distance plots and a wider search. Silhouette is not equally suitable for every cluster geometry. Ground-truth labels are reserved for post-hoc evaluation.

12. Reproducibility

See the accompanying .ipynb for full executable code: data loading, preprocessing, PCA, scree plot, elbow/silhouette selection, DBSCAN parameter scan, hierarchical dendrogram, 2D/3D Seaborn/Matplotlib and Plotly visualizations, metrics, and conclusions. Run all cells top-to-bottom with scikit-learn, pandas, numpy, scipy, matplotlib, seaborn and plotly installed. Random state=42.

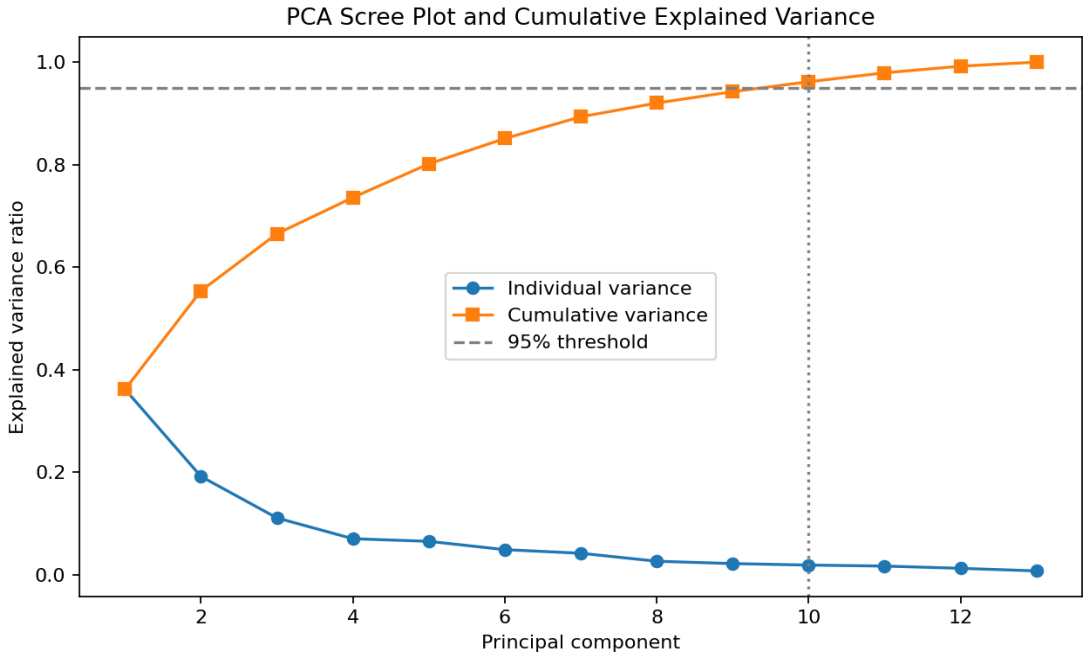


Figure 1. PCA scree and cumulative explained variance.

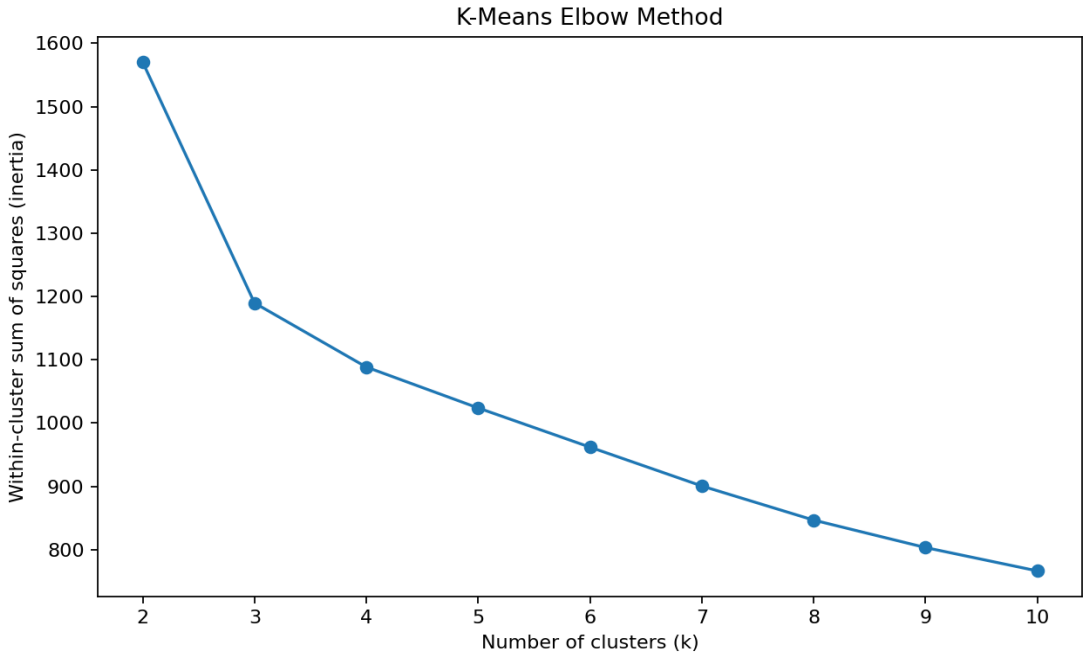


Figure 2. K-Means elbow curve.

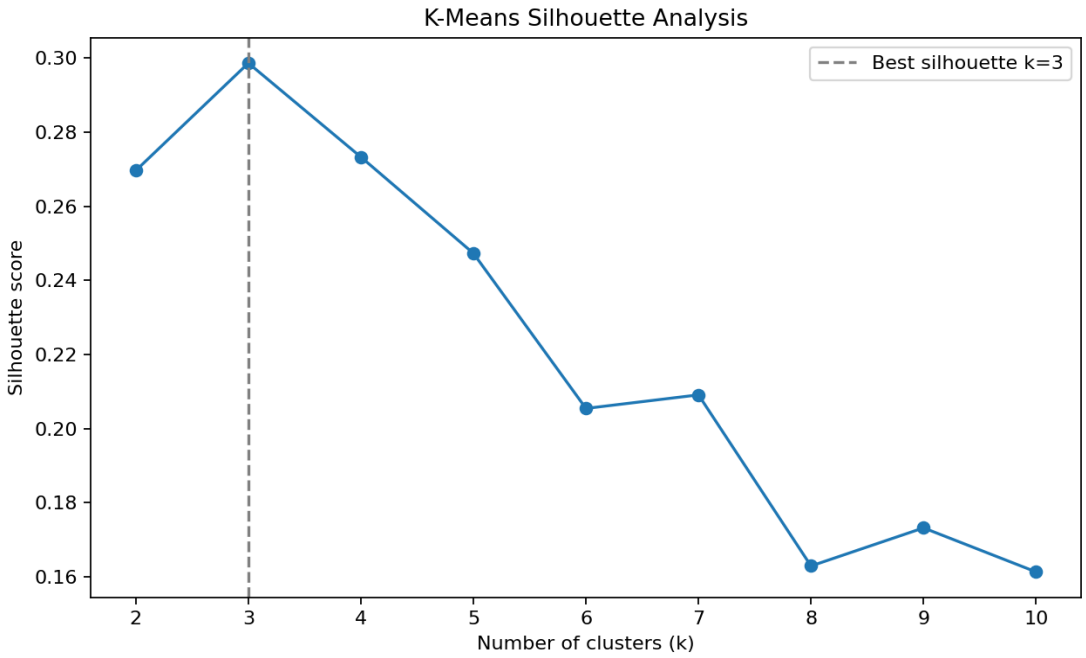


Figure 3. K-Means silhouette analysis.

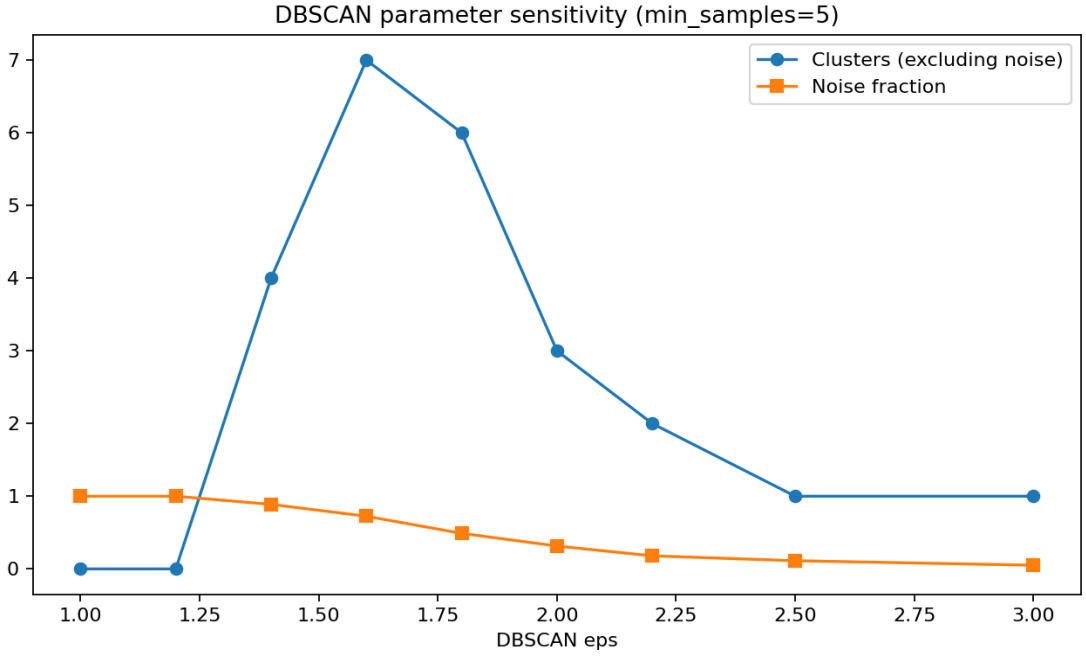


Figure 4. DBSCAN parameter sensitivity.

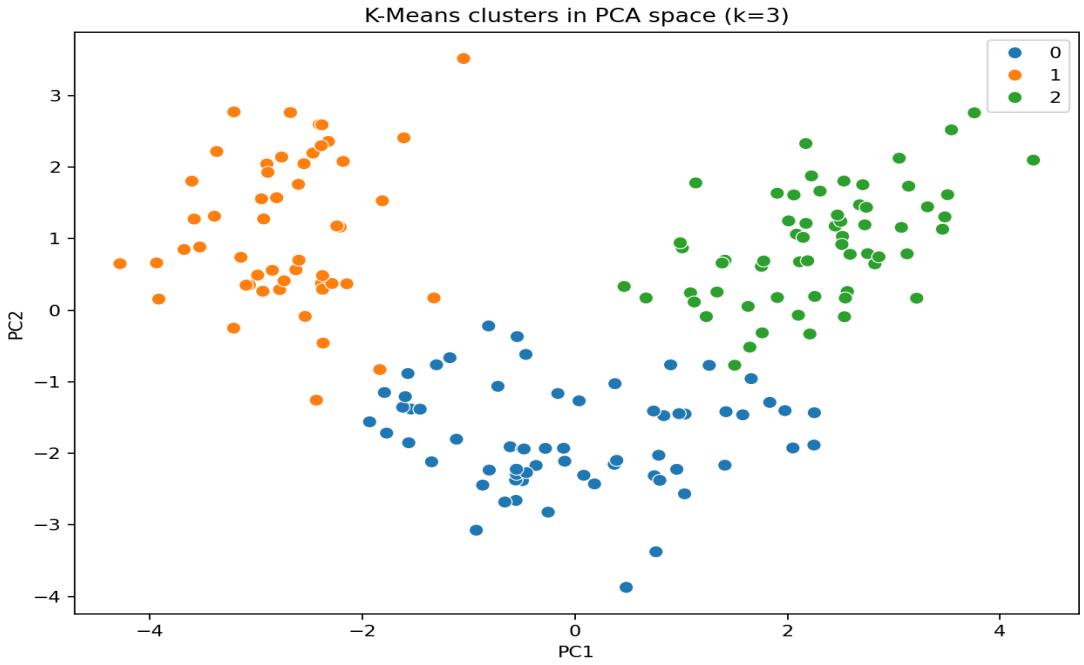


Figure 5. K-Means clusters in two PCA dimensions.

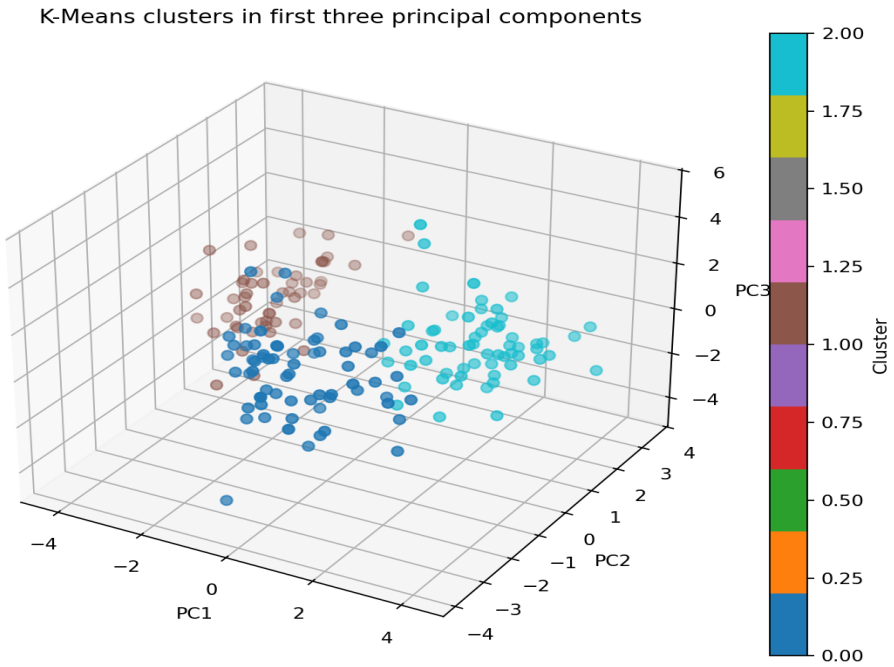


Figure 6. K-Means clusters in three PCA dimensions.